PHASE 2 · DATA SOURCING · VERIFIED 5 SEPTEMBER 2026

Phase 2 datasets

Phase 1's blocker is that CISA KEV is our only exploitation ground truth, and it is incomplete. These are the sources that fix that — checked live for availability, cost, schema and licence rather than recalled.

Companion to **LEV-calibration-Phase1.pdf** · machine-readable version in **DATASETS.md**

The dataset that unblocks the paper is free.

VulnCheck KEV Community needs a free account, not a licence. This was previously assumed to be a commercial purchase; it is not. There is no budget item in Phase 2.

TIER 1 Get these first

1. VulnCheck KEV (Community) — the one that matters

Cost	**Free** — community account at `console.vulncheck.com`, generate a token
Endpoint	`GET https://api.vulncheck.com/v3/backup/vulncheck-kev` with `Authorization: Bearer <token>` — returns a bulk download link
Verified	Returns HTTP 401 without a token, confirming the endpoint exists and needs only the free key
Rate limit	1,000 requests / minute for community users
Coverage	**884 KEVs added in 2026 against CISA's 245** over the same period — roughly 3.6×
Licence	Attribution requested; formal redistribution terms **not published** — confirm before republishing raw data

Why this source and not simply a bigger one. Its schema is close to purpose-built for the estimator we need:

```
cve[], cwes[], vendorProject, product,
date_added                                  <- VulnCheck's own first-report date
cisa_date_added                             <- CISA's, inside the same record
vulncheck_reported_exploitation[]           <- {url, date_added} per reporting source
vulncheck_xdb[]                             <- {xdb_id, exploit_type, date_added}
reported_exploited_by_vulncheck_canaries    <- honeypot-observed (boolean)
```

`cisa_date_added` sitting inside every VulnCheck record means the CISA subset is identifiable **without a join** — the overlap that capture–recapture consumes comes free and unambiguous, with no fuzzy matching to defend in review. `vulncheck_reported_exploitation[]` goes further, naming each independent reporting source with its own date, so the estimator extends to many sources without acquiring them separately. `reported_exploited_by_vulncheck_canaries` is honeypot-observed exploitation — sensor evidence rather than curated advisories, so it is genuinely independent of the process that builds CISA KEV.

2. CISA KEV daily archive — fixes an existing threat, costs nothing

We currently hold *one* snapshot of KEV and trust its `dateAdded` field. Daily archives let us verify that field is never backdated or revised — README threat 17, presently stated but unverified. It also makes the as-of-date KEV rates in `dn_sweep.py` and `plateau.py` genuinely point-in-time rather than reconstructed from a single file.

Source	`github.com/cisagov/kev-data` (official) or `github.com/hrbrmstr/cisa-known-exploited-vulns` (MIT, daily ISO-stamped snapshots)
Cost	**Free**

3. Full daily EPSS archive

Removes the 3.9% monthly-sampling approximation entirely. Not urgent — that error is measured and bounded — but required if any claim ever turns on a difference of that size.

Source	`github.com/empiricalsec/epss_scores` — official historical repository, ~2.5 GB
Cost	**Free**; no licence declared on the repository, worth confirming for redistribution

`TIER 2` Strengthens the independence argument

4. GreyNoise — Community Research Program

Observed scanning and exploitation attempts from sensors. Its independence from KEV is the strongest of any source here: it observes traffic rather than curating advisories, so its blind spots are uncorrelated with CISA's — which is exactly the assumption capture–recapture requires.

Cost	**Free** for non-commercial research — **application required and reviewed**, so apply early
Apply	`info.greynoise.io/community/research-program-request`
Gives	Full Enterprise API including historical data

5. Nuclei templates

A community-written detection template for a CVE is evidence someone judged it worth detecting in the wild. A weaker signal than the others, but MIT-licensed and trivially joinable, and it extends the exploit-availability axis already built from Exploit-DB and Metasploit.

Source	`github.com/projectdiscovery/nuclei-templates`
Licence	**MIT** — the cleanest terms of anything on this list

TIER 3 Optional, low recall

SOURCE	ASSESSMENT
Project Zero 0day In the Wild	~150–200 CVEs, very high precision. Lives in a Google Sheet, not the GitHub repository — checked, the repo holds only root-cause analyses. A gold-standard spot check; far too small to estimate from.
AttackerKB (Rapid7)	Read-only API, but only ~300 CVEs carry usable ratings, and they are human judgement rather than observation.
Shadowserver	Public dashboard needs no account; bulk research data needs an Alliance partnership or reciprocal data-sharing agreement via `contact@shadowserver.org`. Slow to arrange — pursue only if GreyNoise is declined.

CONSEQUENCE

What this changes about the paper

With VulnCheck KEV alone, three things become possible for the first time.

1. Estimate exploitation that **neither** catalog recorded, via stratified capture–recapture — replacing the vendor-ceiling bound whose weakness is already flagged as a limitation.
2. Test whether the over-confidence shrinks against a **broader** ground truth. VulnCheck is ~3.6× larger, so if the 17.2× gap is mostly catalog blindness it should fall sharply; if it barely moves, that is evidence the residual is LEV error. **Either result is publishable**, which is what makes this the right next step rather than a gamble.
3. Use `vulncheck_reported_exploitation[]` to stratify by *reporting source* and test whether the visibility bias found in CISA KEV — 32.5% of Microsoft CVEs against 2.3% of XWiki's, at identical LEV — reproduces in an independent catalog.

> **Do not drop CISA KEV.** The comparison between the two catalogs is the measurement, so both are required. VulnCheck replaces nothing.

OPEN

What could not be verified, and matters before submission

> **Licensing is unresolved on the two most important sources.** VulnCheck publishes no formal redistribution terms — they request prominent attribution, but nothing states whether derived data may be published alongside a paper or archived for reproducibility. The EPSS historical repository declares no licence at all. One email to each settles both, and it should happen before submission rather than after review.

> **Everything above is free.** Verified 5 September 2026. The only resource cost in Phase 2 is roughly 50 GB of disk for the full daily EPSS archive.

Sources checked 5 September 2026 — vulncheck.com/kev · docs.vulncheck.com/community/vulncheck-kev/schema · docs.vulncheck.com/api/v3 · vulncheck.com/blog/state-of-exploitation-2026 · info.greynoise.io/community/research-program-request · github.com/cisagov/kev-data · github.com/empiricalsec/epss_scores · first.org/epss/data · github.com/projectdiscovery/nuclei-templates · googleprojectzero.blogspot.com/p/0day.html